LECTURE 1.5 · EXTENDED READING

Fixed Point and Newton–Raphson

Where x = g(x) and the Newton step come from · Papanikolaou (2026) §2.13, Appendix C

Goes with Lecture 1.5, slide 9: Review of numerical methods

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Extended reading

The root of f is a fixed point of g

1 **Any equation, written as a root**

$$f(x) = 0$$

2 **Subtract it from x: nothing changes, because f(x) = 0**

$$x = x - f(x)$$

3 **Call the right-hand side g, and check it backwards**

$$g(x) \equiv x - f(x)$$

$$x = g(x) \;\Leftrightarrow\; x = x - f(x) \;\Leftrightarrow\; 0 = -f(x) \;\Leftrightarrow\; f(x) = 0$$

Subtract x from both sides: same solutions. The root of f is a fixed point of g.

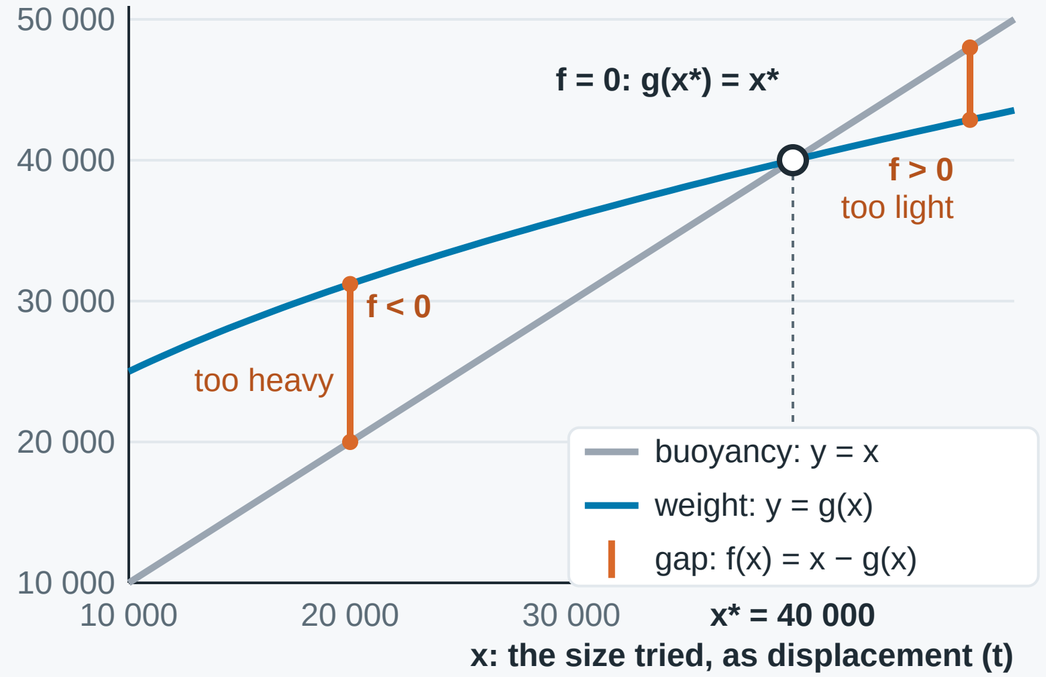
4 **For the ship, x = Δ: f = buoyancy − weight**

$$f(x) = x - [\mathrm{DWT} + W_L(x)]$$

$$g(x) = x - f(x) = \mathrm{DWT} + W_L(x)$$

So g is the weight of the ship of size x, and x = g(x) says buoyancy = weight.

Not the only g: $x = x - c\,f(x),\ c \neq 0$ — c = 1: weight; c = 1/f′: Newton



Try a size x: below x* the ship is too heavy to float at that size, above x* too light. At x* weight = buoyancy.

Extended reading

Newton follows the tangent down to zero

1 **Near the guess x_k, replace f by its tangent**

$$f(x) \approx f(x_k) + f'(x_k)\,(x - x_k)$$

2 **Take the next guess where the tangent is zero**

$$f(x_k) + f'(x_k)\,(x_{k+1} - x_k) = 0 \quad \Rightarrow \quad x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

3 **Also a fixed point: g with c = 1/f′ (slide 2)**

$$g_N(x) \equiv x - \frac{f(x)}{f'(x)}, \qquad f(x^*) = 0 \;\Rightarrow\; g_N(x^*) = x^*$$

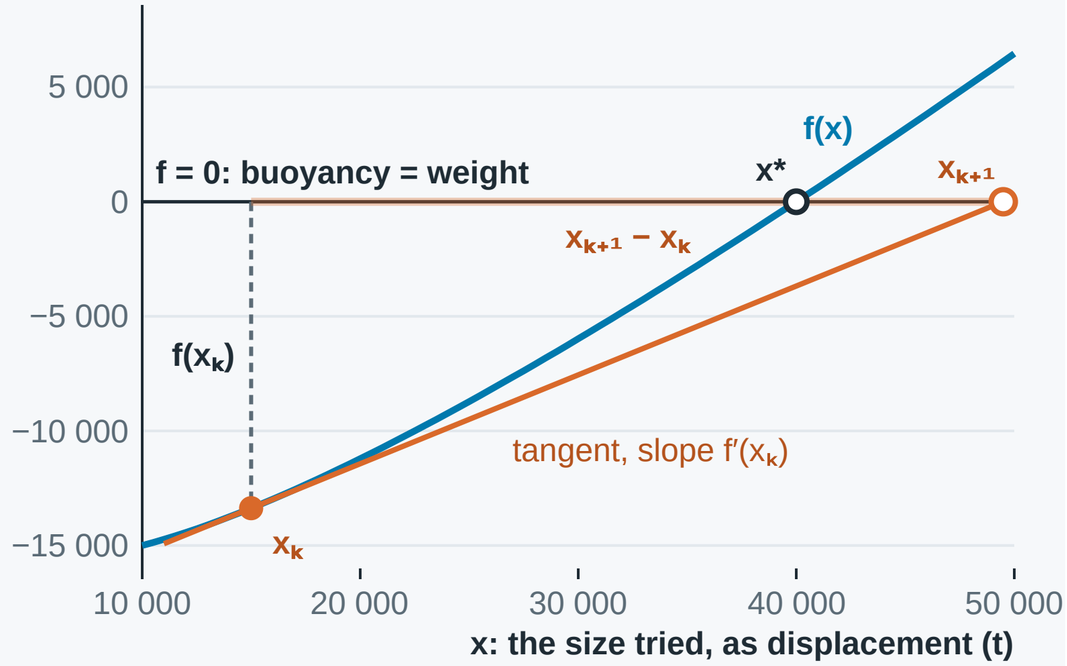
4 **For the ship, x = Δ: f = x − g(x), g the weight**

$$f(x) = x - g(x), \quad f'(x) = 1 - g'(x)$$

$$x_{k+1} = x_k - \frac{x_k - g(x_k)}{1 - g'(x_k)}$$

Step = gap ÷ slope. The fixed point takes the slope as 1.

Why it is fast: $g_N'(x^*) = \dfrac{f(x^*)\,f''(x^*)}{f'(x^*)^2} = 0$ error squared



From x_k, follow the tangent down to zero. Slope = rise ÷ run: $f'(x_k) = -f(x_k) \div (x_{k+1} - x_k)$, which is step 2.